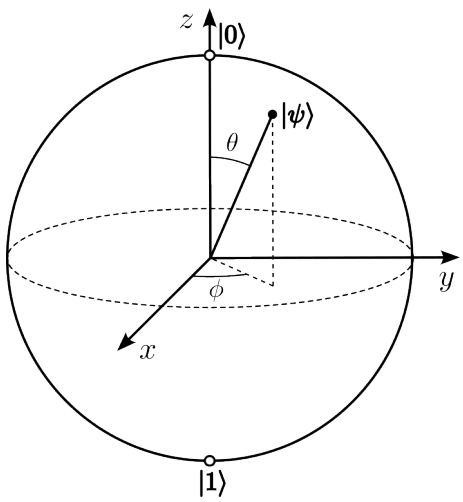

Figure 2.1: The Bloch sphere. Any qubit state on the form (2.34) corresponds to a point on the surface of the unit sphere, parametrised by the polar angle $\theta \in [0, \pi]$ and the azimuthal angle ϕ. The states at the extremal points along each axis x, y, z are the eigenstates of $\hat{\sigma}_x$, $\hat{\sigma}_y$, $\hat{\sigma}_z$. E.g. the states at the 'north and south poles' along z are $|0\rangle$ and $|1\rangle$. Points in the interior correspond to mixed states via (2.75). The origin corresponds to the maximally mixed state.

Here, θ and ϕ can be understood as the spherical coordinates of a point on the surface of the unit sphere in $\mathbb{R}^3$. This provides an intuitive and extremely useful visual representation of qubit states in terms of points on this unit sphere, known as the *Bloch sphere*. It is illustrated in Fig. 2.1. The vector pointing to a given state is known as the *Bloch vector* and can be thought of as a visual representation of the spin of the qubit.

2.2.2 Pauli observables and operators

A set of very important quantities in qubit systems are the Pauli operators or Pauli observables $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$. They correspond to the three spatial components of angular momentum for spin-$\frac{1}{2}$ systems. The Pauli operators are both unitary and Hermitian

$$\hat{\sigma}_k^\dagger \hat{\sigma}_k = \hat{\sigma}_k \hat{\sigma}_k^\dagger = \mathbb{1}, \tag{2.35}$$

$$\hat{\sigma}_k^\dagger = \hat{\sigma}_k, \tag{2.36}$$

for $k = x, y, z$. This means they can be understood both as unitary operations that transform qubit states and as observables that can be measured. It also implies that each Pauli operator is its own inverse, $\hat{\sigma}_k^{-1} = \hat{\sigma}_k$ and hence $\hat{\sigma}_k^2 = \mathbb{1}$ for $k = x, y, z$. They obey the angular-momentum commutation relations

$$[\hat{\sigma}_x, \hat{\sigma}_y] = 2i\hat{\sigma}_z, \tag{2.37}$$

$$[\hat{\sigma}_y, \hat{\sigma}_z] = 2i\hat{\sigma}_x, \tag{2.38}$$

$$[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y, \tag{2.39}$$

where $[A, B] = AB - BA$ denotes the commutator. This can be written compactly as $[\hat{\sigma}_k, \hat{\sigma}_l] = 2i \sum_m \varepsilon_{klm} \hat{\sigma}_m$, where ε_{klm} is the Levi-Cevita symbol. They also anti-commute

$$\{\hat{\sigma}_k, \hat{\sigma}_l\} = \delta_{kl} \mathbb{1}, \tag{2.40}$$

where $\{A, B\} = AB + BA$ denotes the anti-commutator.

Paulis as observables

Let us now first consider the Pauli operators as observables. What are the possible outcomes of a Pauli measurement? Because $\hat{\sigma}_k^2 = \mathbb{1}$, the eigenvalues of a Pauli operator are ± 1. These are then the possible outcomes of any measurement of $\hat{\sigma}_x$, $\hat{\sigma}_y$, or $\hat{\sigma}_z$. We take the canonical basis states $|0\rangle$ and $|1\rangle$ to be the $+1$ and -1 eigenstates of $\hat{\sigma}_z$, respectively. On the Bloch sphere, they sit at the 'north' and 'south' poles along the z-axis, as seen in Fig. 2.1. The eigenstates of $\hat{\sigma}_x$ are

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle), \tag{2.41}$$

and the eigenstates of $\hat{\sigma}_y$ are

$$|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle). \tag{2.42}$$

On the Bloch sphere, they sit at the 'equator' along the x- and y-axes. Expressed as matrices in the eigenbasis of $\hat{\sigma}_z$, the Pauli operators are

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.43}$$

A measurement of $\hat{\sigma}_z$ is projective with projectors

$$\hat{P}_+^z = |0\rangle\langle 0| \qquad \text{and} \qquad \hat{P}_-^z = |1\rangle\langle 1|, \tag{2.44}$$

corresponding to outcomes ± 1. The observable can be written

$$\hat{\sigma}_z = \hat{P}_+^z - \hat{P}_-^z = |0\rangle\langle 0| - |1\rangle\langle 1|. \tag{2.45}$$

From the form (2.34), we see that the probabilities for obtaining each outcome for a generic state is $p(+1) = |\langle 0|\psi\rangle|^2 = \cos^2(\theta/2)$ and $p(-1) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta/2)$. Using $\cos(\theta) = 2\cos^2(\theta/2) - 1 = 1 - 2\sin^2(\theta/2)$ we see that the probabilities are exactly $p(\pm 1) = \frac{1}{2}(1 \pm r_z)$, where r_z is the projection of the Bloch vector onto the z-axis. Similarly,

$$\hat{\sigma}_x = \hat{P}_+^x - \hat{P}_-^x = |+\rangle\langle +| - |-\rangle\langle -|, \tag{2.46}$$

$$\hat{\sigma}_y = \hat{P}_+^y - \hat{P}_-^y = |+i\rangle\langle +i| - |-i\rangle\langle -i|, \tag{2.47}$$

and the outcome probabilities are related to the projections of the Bloch vector onto the x- and y-axes in the same manner. In particular, we note than when the system is in an eigenstate of one of the Pauli operators then the outcome of a measurement of any of the other Pauli operators will be completely random, $p(+1) = p(-1) = \frac{1}{2}$.

Paulis as gates

Next, let us consider the Pauli operators as unitary transformations. What do they do to a qubit? One way to understand them is as logical operations (or 'gates'). Using (2.41), we can rewrite $\hat{\sigma}_x$ as

$$\hat{\sigma}_x = |0\rangle\langle 1| + |1\rangle\langle 0|. \tag{2.48}$$

From here, it is easy to see that the effect of $\hat{\sigma}_x$ on the computational basis states is to flip them

$$\hat{\sigma}_x|0\rangle = |1\rangle, \tag{2.49}$$
$$\hat{\sigma}_x|1\rangle = |0\rangle. \tag{2.50}$$

Thus, $\hat{\sigma}_x$ implements a *bit flip* on the qubit. Similarly, since by definition the computational basis is the eigenbasis of $\hat{\sigma}_z$, we have

$$\hat{\sigma}_z|0\rangle = |0\rangle, \tag{2.51}$$
$$\hat{\sigma}_z|1\rangle = -|1\rangle. \tag{2.52}$$

Therefore, $\hat{\sigma}_z$ implements a *phase flip*. In other words, it flips the relative phase between the computational basis states, e.g. $\hat{\sigma}_z|+\rangle = |-\rangle$. Since, from the commutation relations, $\hat{\sigma}_y = i\hat{\sigma}_x\hat{\sigma}_z$ the effect of $\hat{\sigma}_y$ is a combined phase and bit flip.

Paulis as rotations

Another way to understand the Pauli operators is a generators of rotations. Since $\hat{\sigma}_x$, $\hat{\sigma}_y$, $\hat{\sigma}_z$ are the angular momentum operators corresponding to each axis x, y, z, they generate rotations around these axes, respectively. More precisely, they define rotation operators

$$\hat{R}_k(\phi) = \exp\left(-i\frac{\phi}{2}\hat{\sigma}_k\right), \qquad k = x, y, z, \tag{2.53}$$

where the matrix exponential on the right-hand side is defined by the series $\exp\left(\hat{A}\right) = \sum_n \frac{1}{n!}\hat{A}^n$. The effect of $\hat{R}_k(\phi)$ acting on a qubit state is to rotate the Bloch vector about the axis k by an angle ϕ. Furthermore, using that $\hat{\sigma}_k^2 = \mathbb{1}$ one can show that

$$\hat{R}_k(\phi) = \cos\left(\frac{\phi}{2}\right)\mathbb{1} - i\sin\left(\frac{\phi}{2}\right)\hat{\sigma}_k. \tag{2.54}$$

In particular, we have $\hat{R}_k(\pi) = -i\hat{\sigma}_k$ so (up to a global phase factor) each Pauli operator corresponds to a rotation by π on the Bloch sphere around the corresponding axis.

The above expressions can be generalised to rotations about arbitrary axes. It is convenient to collect the Pauli operators into a vector $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$. For any unit vector $\mathbf{n}$ we can then define

$$\hat{R}_{\mathbf{n}}(\phi) = \exp\left(-i\frac{\phi}{2}\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}\right), \tag{2.55}$$

where $\mathbf{n} \cdot \hat{\boldsymbol{\sigma}} = n_x \hat{\sigma}_x + n_y \hat{\sigma}_y + n_z \hat{\sigma}_z$. This operator corresponds to a rotation by ϕ around an axis pointing along $\mathbf{n}$. Similarly to above, we also have

$$\hat{R}_{\mathbf{n}}(\phi) = \cos\left(\frac{\phi}{2}\right)\mathbb{1} - i\sin\left(\frac{\phi}{2}\right)\mathbf{n} \cdot \hat{\boldsymbol{\sigma}}. \tag{2.56}$$

2.3 Entanglement and density operators

As soon as we consider composite systems – systems made up of two or more parts – we open up for correlations among the subsystems. Quantum systems can be correlated in new ways that classical systems cannot. It is also very often the case that a system of interest is actually a subsystem of some larger system which we are not interested in or unable to describe in detail. For example, a qubit implemented in some physical system, say an atom, cannot generally be perfectly isolated from the environment. Therefore, we need ways to describe states, evolution, and measurements of parts of systems. The axioms in Sec. 2.1 are give a perfectly valid and complete formulation of quantum mechanics, but they assume a closed system. In a sense, they describe the entire universe. When considering only a small part of the universe, i.e. a subsystem, they need to be revised. In particular, in general for subsystems, states are *not* rays and evolution is *not* unitary.

2.3.1 Entanglement for pure states

We start by considering quantum correlations when the total system is closed (so the axioms from before apply and its state is still a ray, i.e. a pure state). As explained in Sec. 2.1.6, the state space of a bipartite system is a product $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. If the states of subsystems A and B are independent, then the state of the total system is also a product

$$|\psi\rangle_{AB} = |\psi_A\rangle_A \otimes |\psi_B\rangle_B. \tag{2.57}$$

Not all bipartite states (2.32) can be factorised like this. The ones that do not factorise necessarily contain some quantum correlations. Any state of a bipartite system which can be written on this product form, for some choice of states $|\psi_A\rangle_A$ on A and $|\psi_B\rangle_B$ on B, is said to be *separable*. Any state which is not separable is said to be *entangled*.

2.3.2 Schmidt decomposition and Schmidt number

For any bipartite pure state $|\psi\rangle_{AB}$, it is always possible to find orthonormal bases $\{|n\rangle_A\}$ and $\{|n'\rangle_B\}$ for A and B, respectively, such that the state can be written

$$|\psi\rangle_{AB} = \sum_n \sqrt{p_n}|n\rangle_A|n'\rangle_B, \tag{2.58}$$

with real, nonnegative coefficients, $p_n \geq 0$. This is known as the *Schmidt decomposition* or *Schimdt form*.

The *Schmidt number* of a state is the number of nonzero coefficients in the Schmidt decomposition. All separable pure states are product states and have a Schmidt number of 1. A Schmidt number larger than 1 implies that the state is entangled.